

Subject Description Form

Subject Code	EIE4413
Subject Title	Digital Signal Processing
Credit Value	3
Level	4
Pre-requisite	EIE3312 Linear Systems
Co-requisite/ Exclusion	Nil
Objectives	This subject aims to provide students with the fundamental concepts and practical know-how in digital signal processing (DSP), important to many communications and multimedia applications. Both theory and practical realisation are stressed.
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Understand the basic concepts of Fourier analysis of digital signals and apply them to practical problems. 2. Design and realize simple digital filters for practical applications. 3. Understand the importance of random signal processing, and its application in statistical measures, prediction, and data modelling. 4. Design and simulate simple DSP systems. <u>Category B: Attributes for all-roundedness</u> 5. Think critically. 6. Learn independently.
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Introduction</u> 1.1 Why DSP? Typical DSP system. Typical steps to construct a DSP system. 2. <u>Discrete Fourier Transform and Convolution</u> 2.1 Fourier series and continuous-time Fourier transform, Gibbs phenomenon, Shannon sampling theorem. Discrete Fourier transform (DFT), properties of DFT, Fourier analysis using DFT. The fast Fourier transform (FFT) algorithm. 2.2 DSP systems. Linear convolution and its implementation. Convolution theorem. Convolution by section. 2.3 The Z transform. Properties. Relationship with the Fourier transform. Pole and zero in the Z domain. 3. <u>Design of Finite Impulse-response (FIR) and Infinite Impulse-response (IIR) Digital Filters</u> 3.1 Design stages for FIR filters. Design method – Windowing. Designing low-pass, band-pass, and high-pass FIR filters. Linear phase response filters and their design. 3.2 Difference equation, impulse response, and transfer function of IIR filters. IIR filter implementation. Poles, zeros, and stability of IIR filters. Frequency

	response of IIR filters. Case study: first and second-order IIR filter design. Designing higher-order IIR filters. 4. <u>Random Signal Processing</u> 4.1 Revision on Random Processes, probability distribution function, expected values, variance, and standard deviation. Application – Finding correlation: covariance, cross correlation, unbiased cross correlation, auto-correlation. Application – Denoising: white and coloured noises, power spectral density, periodogram, Wiener filter, Welch’s method. 5. <u>Adaptive Filters</u> 5.1 Correlation canceller. Using the Wiener filter for correlation cancelling. Steepest descent algorithm. Basic Widrow-Hopf LMS algorithm. Application – Active noise cancellation. Laboratory Experiments: 1. Python for DSP laboratory exercises. 2. Properties of DFT and the fast Fourier transform. 3. FIR filter analysis and design. 4. Statistical digital signal processing.		
Teaching/ Learning Methodology	Teaching and Learning Method	Intended Subject Learning Outcome	Remarks
	Lectures	1, 2, 3, 5	Fundamental principles and key concepts of the subject are delivered to students
	Tutorials	1, 2, 3, 5	Supplementary to lectures, tutorials are conducted with a smaller class size. Students will be able to clarify concepts and have a deeper understanding of the lecture material; problems and application examples are given and discussed.
	Laboratory sessions	1, 2, 3, 4, 5, 6	Students will make use of the software tool to simulate the various theories and visualize the results.

**Assessment Methods
in Alignment of
Assessment and
Intended Subject
Learning Outcomes**

Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)					
		1	2	3	4	5	6
1. Continuous Assessment (total 40%)							
• Short exercises	5%	✓	✓	✓		✓	
• Tests	20%	✓	✓	✓		✓	
• HW Assignment	5%	✓	✓	✓		✓	✓
• Laboratory sessions	10%	✓	✓	✓	✓	✓	✓
2. Examination	60%	✓	✓	✓		✓	
Total	100%						

The continuous assessment includes assignments, laboratory reports, short exercises, and two tests.

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Specific Assessment Methods/Tasks	Remark
Short exercises	Small exercises are conducted to measure students' basic understanding of the theories, concepts, and physical meanings of subject materials during the lectures or tutorial classes.
Test and examination	End-of-chapter type problems are used to evaluate students' ability in applying concepts and skills learnt in the classroom, and their comprehension of subject materials. Students need to think critically to devise solutions to the problems given.
Assignment	Students should learn independently the related concepts and search for the required information and data to formulate the solution.
Laboratory sessions	Each student is required to produce a report on the laboratory work they conduct. Each student also needs to make a demonstration on the open-ended question set out in each laboratory work.

Student Study Effort Expected	Class contact (time-tabled):	
	• Lecture	26 Hours
	• Tutorial/Laboratory/Practice Classes	13 Hours
	Other students' study effort:	
	• Lecture: preview/review of notes; homework/assignment; preparation for tests/examination	36 Hours
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision, and/or report writing	30 Hours
	Total student study effort:	105 Hours
Reading List and References	Textbook: 1. R. Anand, <i>Digital Signal Processing: An Introduction</i>, Dulles, Virginia: Mercury Learning and Information, 2022.. Reference Books: 1. B. Maurice and E.A. Benjamin, <i>Digital Signal Processing: Theory and Practice</i>. Newark: John Wiley & Sons, 10th ed., 2024. 2. S.K. Mitra, <i>Digital Signal Processing</i>, McGraw-Hill Education (Asia), 3rd ed., 2009.	

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